

My Little Brain

It knows you

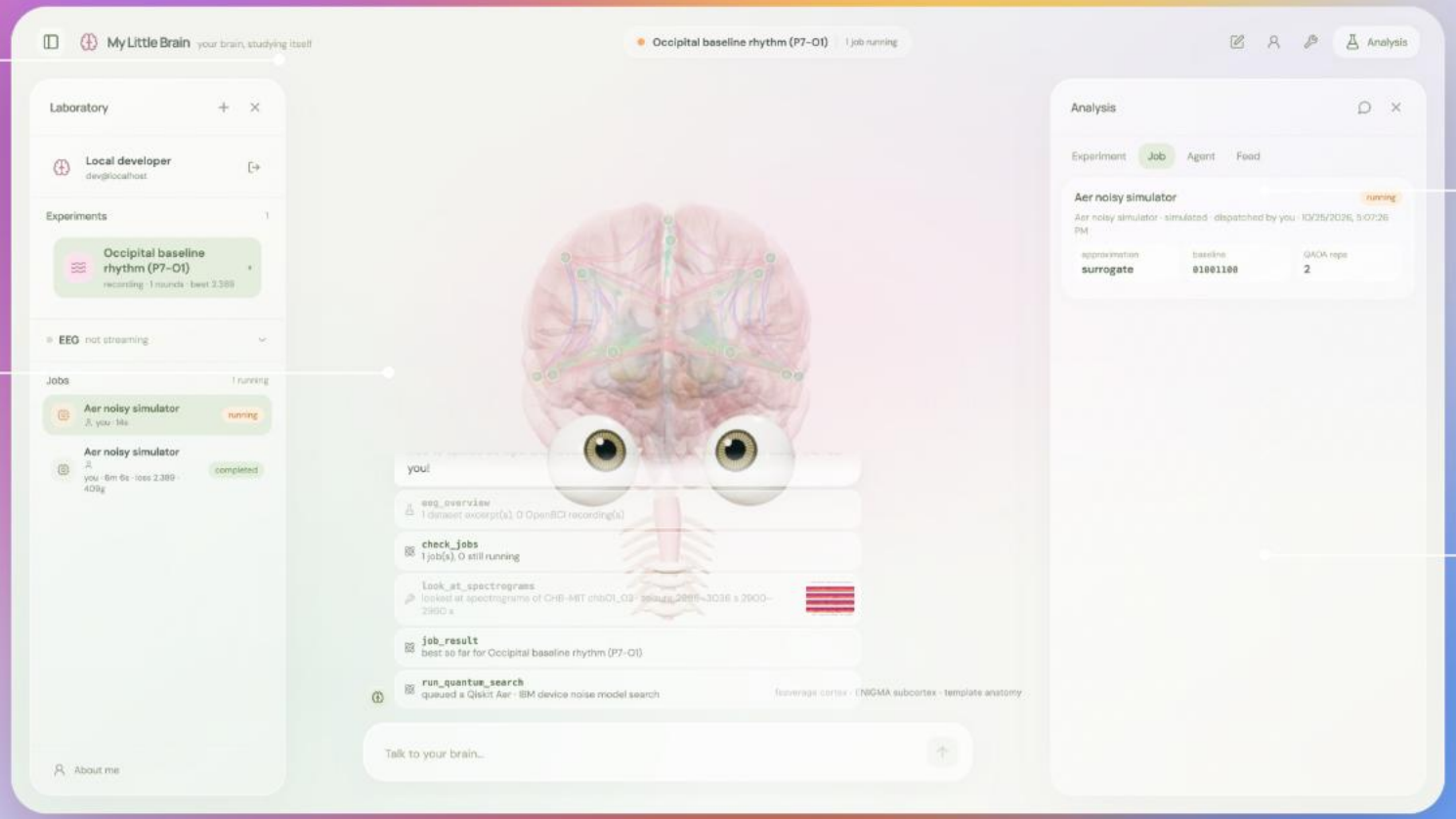
your notes · your live EEG · your questions

it watches your live EEG

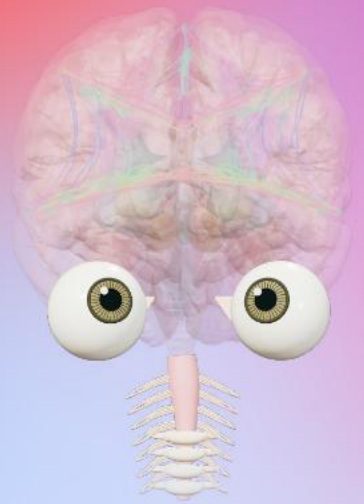
you chat with your brain

it designs quantum circuits

results land here



It studies itself



IONQ
IonQ · trapped ions

IBM
IBM · heavy-hex

qBraid
qBraid · Rigetti/IQM/AQT

Extropic · THRML

Gemini · vision + agent

OpenBCI · your EEG

48 / 256
simulations to recover a hidden brain state

42 gates
a circuit Gemini wrote itself (from 106)

52 vs 454
gates on IonQ vs IBM, same problem

τ 16→55 ms
seizure windows point to slower synapses

OpenQASM written by Gemini 4 Argon · Qiskit Aer, IBM Torino noise · job f8fdb905…

```
OPENQASM 2.0;
…
qreg q[8];
creg meas[8];
ry(1.7) q[7];
x q[6];
x q[5];
x q[2];
ry(2.5) q[1];
cx q[7],q[6];
cx q[6],q[4];
cx q[7],q[3];
ry(-0.55) q[3];
…
```

model hypotheses, not diagnoses · no quantum-advantage claim

EEG → quantum state

Your EEG → 10 numbers

relative power in 10 frequency bands, per channel

4 brain parameters × 2 bits = 8 qubits

drive · inhibition · excitation · synaptic time constant

8 qubits = a 256-dimensional Hilbert space

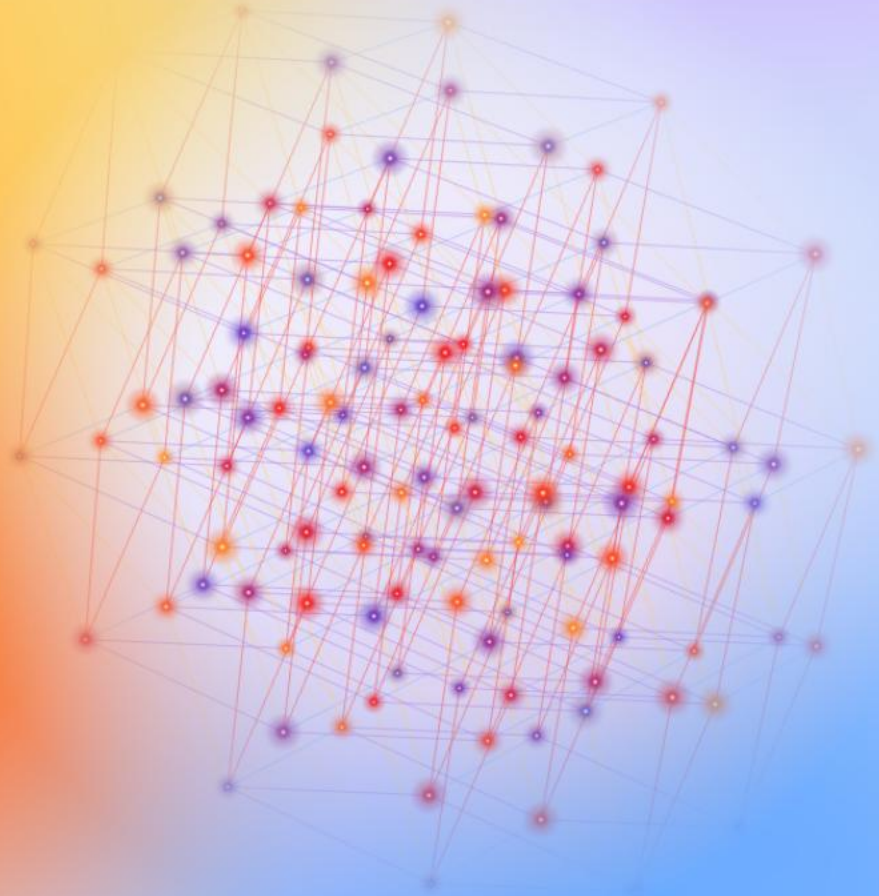
every basis state is one possible brain: $|\psi\rangle = \Sigma\ \alpha_b\ |b\rangle$

Your EEG shapes the circuit

$\|f(\theta) - y\|^2$ → Ising energy → QAOA amplifies the best-fitting brains

Measure → simulate → verify

top samples are re-simulated; nothing is trusted unverified



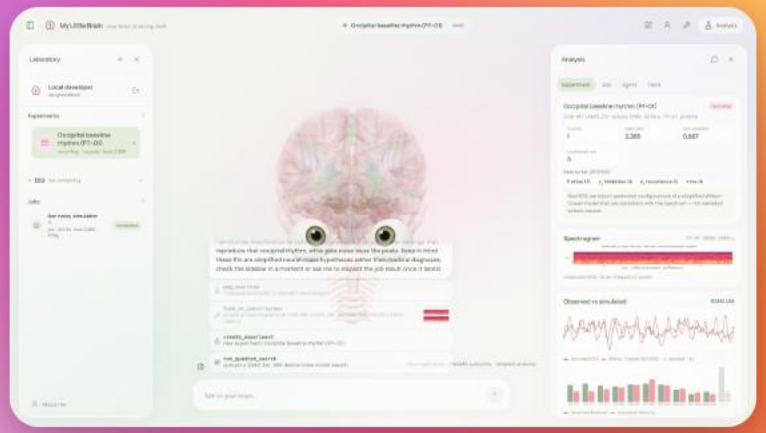
+1 bit per parameter → 12 qubits, 4,096 brains · each qubit doubles the space · today's 256 are classically enumerable

Behind the scenes



OpenBCI Ultracortex

Web Serial · Chrome



The app

chat · 3D brain · experiments

Gemini 4 Argon

agent · Cloud Run

GOOGLE CLOUD

Cloud Run · Firestore · Storage · Cloud Tasks

Secrets · Registry · Sign-In

QAOA · thermodynamic sampling

IONQ

IBM

qBraid

Qiskit Aer

Extropic THRML

validated by forward simulation

4 brain parameters × 2 bits = 8 qubits = 256 hypotheses · nonlinear loss → QUBO surrogate → exact Ising map

The team



Jacob Valdez

Member of Technical Staff, Stealth Startup
jvboid.dev



Hanah Rahman

Student Researcher, Berkeley Lab
quantum applications · quantum ML



Rohit Korrapolu

Applied AI researcher
i14labs